Deploying Generative AI LLMs on Docker

Built on massive datasets, large language models or LLMs are closely associated with generative AI. Integrating these models with Docker has quite a few advantages.

Figure 1: Pretrained robotic deployments for AI applications

Generative AI, a growing and prominent segment of artificial intelligence, refers to systems capable of producing content autonomously, ranging from text, images and music, to even code. Unlike traditional AI systems, which are primarily deterministic and perform tasks based on explicit rules or supervised learning, generative AI models are designed to create new data that mirrors the characteristics of their training data. This capability has profound implications, transforming industries by automating creative processes, enhancing human creativity, and opening new avenues for innovation.

According to *Fortune Business Insights*, the global generative AI market was valued at approximately US$ 43.87 billion in 2023. This market is projected to increase from US$ 67.18 billion in 2024 to US$ 967.65 billion by 2032. This dominance can be attributed to several factors, including the presence of key technology companies, significant investments in AI research and development, and a robust ecosystem that fosters innovation and collaboration. As businesses across various sectors increasingly adopt generative AI solutions to enhance their operations, the market is poised for unprecedented growth and transformation in the coming years.

Large language models (LLMs) are closely associated with generative AI and specifically focused on text generation and comprehension. LLMs such as OpenAI's GPT-4 and Google's PaLM are built on massive datasets encompassing a wide range of human knowledge. These models are trained to understand and generate human language with a high degree of coherence and fluency, making them instrumental in applications ranging from conversational agents to automated content creation.

The versatility of LLMs is evident in their application across various domains. In the healthcare industry, LLMs are used to assist in the drafting of clinical notes and patient communication. In finance, they generate reports and assist with customer service. The adaptability of these models makes them a cornerstone of the AI revolution, driving innovation across multiple sectors.

The deployment of large language models (LLMs) involves intricate processes that require not only advanced computational resources but also sophisticated platforms to ensure efficient, scalable, and secure operations. With the rise of LLMs in various domains, selecting the appropriate platform for deployment is crucial for maximising the potential of these models while balancing cost, performance, and flexibility.